

EE 314 : Digital Signal Processing

Course Code : EE 314

Course Name : Digital Signal Processing

L-T-P-C : 3-0-2-4

Intended for : B.Tech. 2nd, 3rd year students

Prerequisite : IC210: Probability, Statistics and Random Processes, EE 304: Communication Theory or the instructor's consent

Mutual Exclusion: EEXXX (Signals & Systems)

Revision:

Approval: 54th BoA

Course Contents

- **Review of discrete-time signals and systems:** LTI systems and their properties, impulse response and convolution, Difference equations, Eigen-functions of LTI systems. (4 hours)
- **Fourier and Z-transforms:** Discrete-time Fourier transform and its properties; Z-transform and its properties; Discrete Fourier Transform and its properties; Fast Fourier Transform. (10 hours)
- **Sampling:** Time domain and frequency domain representation, Nyquist theorem, Signal reconstruction, Discrete-time processing of continuous-time signals, Continuous-time processing of discrete-time signals, Changing the sampling rate, Multi-rate signal processing, Sub-Nyquist sampling and its applications. (8 hours)
- **Filtering:** Discrete-time frequency selective filtering, Phase distortion and delay, Characterization with difference equations, Stability and Causality, Frequency response of rational system functions, All pass and minimum phase systems, Basics of filter design, Z-transform characterization of IIR filters, Window functions for FIR filters, Filter structures for IIR and FIR filters. (10 hours)
- **Introduction to wavelets and their applications:** Haar expansions, Wavelets in continuous time, Discrete Wavelet Transform using Haar basis, Construction of wavelets using Fourier techniques, Multiresolution analysis and construction of the wavelet. (10 hours)

Laboratory/practical/tutorial Modules:

- The exercises/projects should cover various real-world applications of DSP. A few suggested topics: image processing and compression; Speech processing; Compressed sensing; Signal processing in communication systems

Textbooks:

1. Alan V. Oppenheim, Ronald W. Schaffer, John R. Buck., **Discrete-Time Signal Processing**, 2nd edition, Pearson, 1999.
2. Martin Vetterli, Jelena Kovacevic and Vivek Goyal, **Fourier and Wavelet Signal Processing**, Cambridge University Press, 2014



References:

1. John G. Proakis, Dimitris G. Manolakis., **Digital Signal Processing – Principles, Algorithms, and Applications**, 4th Edition, Pearson 2007.
2. Stephen Mallat, **A Wavelet Tour of Signal Processing The Sparse Way**, Elsevier, 2009